

Calibrating an enzyme- and temperature-constrained model: what
was learned, what was corrected, and what is still open

The record from P16 to T3, assembled 2026-09-16

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1 How to read this document

This is the successor to `reports/synthesis/` for everything after 9 September 2026. It does not replace that document: the synthesis remains the record of the framework, the seven organism models, the Candida results and the audits, and its §1–§6 stand. What it could not do was answer the question it ended on — whether a posterior was reachable for the *E. coli* gas-exchange calibration. That question has now been worked for a week, and this is what came of it.

Every number here is read from a committed file, and the file and the commit are given. Where reports/synthesis/evidence.csv carries a row for a claim, it is cited by row id like [T2b]. Where it does not — P9 through P17 predate or postdate that file’s rows — the file and commit are named inline. Nothing is quoted from memory or from another summary.

The document is organised by what was learned, not by the order the prompts ran. Seven things were established, in a sequence where each one was only visible once the previous had been fixed. That order is the argument.

Self-corrections appear with the same prominence as everything else, and there are several. A multimodality reading was withdrawn; a sampleability criterion was found to contain a conjunct that tested the wrong failure mode; a parameter was discovered never to have been read by the model; a “96 %” was corrected to 16.5 %; a remedy was measured and found worse than doing nothing. Two of these changed a conclusion. The even-handedness is not modesty — it is the reason the remaining claims can be trusted.

One thing to hold throughout: the framework works, is gated, and is not in question. What is unsolved is the *calibration* of one configuration of one organism, and its obstacle now has a name and a price.

2 Where it stood

The synthesis of 9 September left three things. A merged framework: one core, seven organism models, two regression gates — the *Candida* gate at **79 of 79 comparisons** and the *E. coli* gas-exchange port at **60 of 60** (reports/candida_thermal_limit/, reports/P1_parsa_port/gate.md @ fa1fa42). A set of parameter medians carrying an explicit *not-converged* label. And no intervals at all.

The reason for the label was already established and is worth restating, because everything that follows is a consequence of it. **No chain in the family had converged — neither the nine refits nor the six inherited ones** — with autocorrelation times of 146–245 against chains of 1500–2000 steps, i.e. chain/ 6–12 against a criterion of 40 (reports/P4_refit/ @ f7abf1f). Adding walkers did not fix it: at 128 walkers over 1500 steps grew **26 → 141** and the verdict was NOT MIXING (reports/P7_walkers/). There was no ridge to blame — the ensemble was isotropic and unimodal, the first principal component carrying only **16.5 %** of the variance with between 110 and 121 on every one of the sixteen components (reports/P8_ridge/).

So the sampler was not obviously weak, the geometry was not obviously hard, and the chains did not mix. The question was why.

3 The likelihood was cliffed, and the sampler was not weak

The answer was that nobody had looked at the surface. When it was scanned — 22 line scans through the working point, each of 41 evaluations at 0.05 posterior standard deviations — **12 of the 22 lines had a single step exceeding 20 % of the line’s entire range**, with a median largest step of **39 %** and a maximum of **99 %** (reports/P9_surface/report.md @ dbb2e3d). The surface was not rough in the sense of noise. It was cliffed.

The mechanism was traced to one place. At cold temperatures the linear program’s optimal oxygen uptake jumps between vertices while growth barely moves, and the respiration term — a Normal in log O with a small variance — turns a two- to four-fold jump in O into tens of log-likelihood units. Tightening solver tolerances did not remove it, so it was not numerical noise: it was the model’s own piecewise-linear structure being scored on a scale that could not absorb it.

Two changes followed, each implemented as a core option, default off, and turned on only for *E. coli* (strains/eciML1515/gas_exchange.yaml @ dea5535 [P13a]):

- **A parsimonious tie-break.** Growth is held at its optimum and total absolute flux minimised, so the oxygen flux read off is a function of the model rather than of the solver’s path.

- **A floor on the log-O standard deviation**, set to the largest vertex jump measured anywhere in the scan, **1.4216** — moved there from an earlier 0.76 which was the largest jump at the *modal* cliff temperature only (reports/P11_nested/report.md @ 4f4c56f).

Together these cut the three decomposed cliffs by about an order of magnitude, from **−70 / −32 / +23** log-likelihood units to **−6 / −3 / +2** (reports/P10_respiration_likelihood/report.md @ 409d70b). The cost is stated where the option is set: the respiration standard deviation is now at least 1.42 in log at every temperature — a factor-four band on O — and the three cold points whose measured relative standard deviation is 0.06–0.09 lose the most. That is the price of a sampleable surface, and it was paid deliberately.

3.1 A criterion that tested the wrong failure mode

The first self-correction belongs here. The rule used to decide whether the surface was sampleable was *relative* — a step above 20 % of a line’s range — and the fixes above shrank the ranges along with the cliffs, so the rule became centre-dependent: the same surface read SMOOTH at one point and ROUGH at another. It was replaced by an **absolute** rule, in log-likelihood units, written before the re-scan; under it the largest step was **3.53** against a 5-unit bar (reports/P11_nested/report.md @ 4f4c56f).

That rule was then itself found to be wrong, in a more interesting way. It named **plateaus** as nested sampling’s failure mode — correctly, since a plateau puts an atom in the distribution of the likelihood under the prior and voids the volume estimate — and then *also* required no jumps. Those are different objects and only one is fatal: a jump leaves a gap in the likelihood’s support, carrying no prior mass, which nothing ever samples. The surplus conjunct cost one run slot and blocked a second, **on a surface that had no plateaus at all**: 0 of 480 adjacent evaluations were exactly equal (reports/P14_posterior/ @ 030c4cf [P14b]).

The instrument that settled it is worth keeping. A step size confounds a steep smooth gradient with a genuine jump; refining the grid separates them in four evaluations, because for a bounded derivative $|\Delta|$ halves with the spacing and for a jump it converges to the jump height. The step that had stopped the work, 8.5517 units, refines **8.5517 → 4.3325 → 2.2295 → 1.1236** — ratios of about one half — and is pure steepness [P14a].

4 One live basin, not many

Meanwhile two nested runs on the cliffed-then-floored surface disagreed by **6.8 sigma in log evidence** (−22.886 against −24.567; reports/P11_nested/report.md @ 4f4c56f), and the natural reading was multimodality — the more so because the published precedent, Pettersen & Almaas’s audit of Li et al.’s yeast etcGEM, reported seed-dependent posteriors and multimodality with 2,292 per-enzyme parameters where we had 16 global ones.

That reading was tested directly and withdrawn. With endpoints converged (11 of 12 on tolerance) and basins defined by bottleneck barriers rather than by clustering, there is **one live basin**. The barrier matrix is bimodal at **13.868 against 1.785** elsewhere, with the basin count stable over a factor-7.8 window of the threshold, and the two runs’ best samples — the supposed two modes — are **0.266 apart**, which is noise (reports/P12_modes/report.md @ 4f4c56f).

The only separated basin is a model with **zero predicted growth at every measured temperature**. Hold that; it becomes §6.

Two things follow. The claim that the multimodality is intrinsic to the thermal formulation is **unsupported by our own evidence**, and is withdrawn by dated note in the synthesis with its numbers unedited. And PCA clustering is not a mode-finder: the isotropy §2 reported was real, and it was the *explanation* attached to it, not the geometry, that was wrong.

5 A parameter that was never read

One of the sixteen sampled parameters, the metabolic proteome fraction **f_metab**, had been classified FLAT — the likelihood insensitive to it — and that classification had been used to argue about identifiability.

It was not flat. **It was not connected.** Configuration D selects a coupled growth-law allocation in which the metabolic fraction is computed from the growth rate, and the branch that does so states in its own comment that the sampled argument is ignored; the only other reader is a simplex guard that cannot fire. The coordinate entered the sampled set the day before the superseding law was written, and was switched on 50 minutes after it, with a docstring true of a neighbouring parameter and false of this one. A documentation discrepancy, not a wiring omission (reports/T1_target_revision/DECISIONS.md D2 @ b7ab6e3).

This was then **proven rather than argued**, which is the part worth keeping. The claim is an algebraic invariant: the target with the coordinate removed must equal the target with it present, at any value of it. Registered in advance with a tolerance of 1e-6 and checked at **870 archived points**, the maximum difference was 2.02×10^{-6} , with 0 violations and 0 unresolved solves [T1a]. Removing it is therefore free: no log-likelihood changes and no evidence changes.

The neighbouring parameter `f_maint` is a different case and stays sampled — it sets both the enzyme-pool bound and the maintenance-ATP bound.

6 Infeasibility was being rewarded

Return to the dead basin. A model that predicts no growth anywhere should be scored badly, and it was — on growth. On respiration it was scored **−0.017** against the live basin’s **−13.549**, because at its temperatures the linear program has no feasible solution, the oxygen prediction is missing, and the term simply skipped it. **The dead basin won about 13.5 log-likelihood units by not being scored at all.**

Part of this had already been fixed. A *feasible-but-not-growing* cell still respire for maintenance, and the earlier support form discarded those temperatures via a growth mask; the `clamp` form scores them at full weight and masks only a genuinely missing oxygen value. The third candidate — imputing $O = 0$ where growth stops — was **withdrawn on measurement**: it replaces a real oxygen uptake of 0.525–5.674 with an epsilon and fails the state test by **119.14 log-likelihood units** [P13b, P13a].

The *infeasible* case was untouched, and closing it needed facts rather than a penalty. Those were assembled from the measurement record and the solver, with nothing invented:

- **The observable and its uncertainty.** The likelihood consumes a per-cell oxygen rate, with the replicate standard deviation measured at every one of the twelve temperatures (five replicates each, so the code’s 30 % floor never fires); the measured signal-to-noise ranges from 2.6 to 17.6. **No detection limit is documented anywhere**, so a censored model is not derivable and was not invented (reports/T1_target_revision/DECISIONS.md D3).
- **A structural zero on growth does not imply one on oxygen.** At 50 °C three of five series carry the pipeline’s deliberate no-growth marker and **still respire**, at the same order as the growing series. The data forbid the obvious substitution.
- **What the missing predictions actually are.** Every missing prediction in the audit set was classified by solver status through a registered three-rung retry ladder: **876 points, 10,512 solves, 9,355 STRUCTURAL_ZERO, 0 UNRESOLVED, 0 resolved on retry** [T1b]. They are genuine infeasibilities, not numerical failures.
- **The existing term cannot score a zero prediction at all.** It is Normal in $\log O$, and $\log 0$ is $-\infty$. A lognormal error substitution is not implied by the code’s use of a log scale; it would be a new modelling choice.

The rule adopted was therefore the **limit of the existing term**, not a penalty and not a fitted constant: a parameter set whose linear program is infeasible at any measured temperature has **zero likelihood**. Unresolved solves remain a separate category that is reported, never scored.

6.1 The 765 points, and a number corrected

The same classification produced the sharpest result of the week. Of the **800** validated sampling points retained by the converged run, **765 are infeasible at every one of the twelve temperatures** [T1b]. Their

log-likelihood is the growth term alone, a function of one nuisance parameter with a ceiling of **-18.6825**, and that ceiling sits *above* most of the living points. They carried **81.5 %** of that run’s posterior weight.

And here the anticipated figure was wrong. It had been expected that ~96 % of a defensible prior would be infeasible. Measured directly — 2,000 independent prior draws — the prior rejection rate is **16.45 %**, **Wilson 95 % interval [14.9 %, 18.1 %]**, and the rejection fires at the coldest temperature in **326 of 329** cases [T2b]. The 96 % was the *live-point* fraction of one run at one iteration, which is where the sampler had concentrated — not the prior. About 32 % of the prior is “living” by the usual criterion.

The distinction matters: the dead stratum held four fifths of the posterior weight from a sixth of the prior’s volume, because the scoring rule rewarded it. It was a likelihood defect, not a volume effect, and not biology.

The revision worked. Implemented as core options (default off, gated **79/79** and **60/60** byte-identical, turned on for one strain), re-verified at all 876 audit points — 13 feasible points unchanged to 9.6×10^{-6} against a 1e-6 tolerance, 863 at $-\infty$, 0 unresolved [T2a] — and confirmed on the runs themselves: **every posterior sample is feasible at every temperature**, and 97.6–98.6 % of predictive draws grow above half the measured peak [T2e]. The dead region is gone.

7 The null test, and what it did and did not show

Before any of the target revision, a separate line of work asked a cleaner question: does the sampler recover a prior it is *known* not to be learning from? An inactive coordinate’s posterior must equal its prior, and that is checkable without any biology.

It failed, and the failure reproduced. Independent reconstruction of two converged fifteen-dimensional runs confirmed that both met their stopping condition — log evidence **-26.029894** and **-26.052380** — and that they did **not** reproduce each other: **14 of 15 marginals fail** the agreement rule, and the provably inactive coordinate’s weighted CDF mean is **0.319120** and **0.393715** against a target of 0.5 (`reports/P16_reduced/audit_summary.json @ b8b181b`).

The follow-up localised the mechanism without repairing it: a **60–80-fold mismatch** between group and pooled proposal widths, ancestry collapse in the live population, stratum-dependent trapping, and seven curvature-sensitive axes whose local curvature is scale-dependent at a saved point. Every within-sampler remedy tried was falsified, and the gate was closed **NEGATIVE** by decision (`reports/P17_inactive_prior/ @ 3a5de32`).

Its two limits are carried verbatim, because they bound everything above and below. The curvature findings concern *one non-optimal point*, not proof that all samplers fail or that likelihood structure is the sole cause. The 80–87 % living fractions in its analytical controls are consequences of specified toy priors and likelihoods — **never a biological target or an acceptance threshold**. And fixing the uniform melting-temperature shift at zero, which both the converged runs and everything after them do, **assumes the meltome’s mean is exact**; any uniform error in it is absorbed by the stability scale and the catalytic parameters.

One result from that work has to be kept in view for §9: **its analytical controls failed with no linear program present at all**. On a narrow-mixture target with an exactly known answer, the active-region probability came out **0.084 / 0.785 / 0.231** against the exact **0.30437**. Whatever else is wrong, a genuine sampler weakness exists independently of the model.

8 The likelihood is not a function of its parameters

With the target revised, gated and verified, the protocol’s five independent runs were launched on reserved seeds. Two finished. The third crashed.

The crash was diagnosed rather than retried. The sampler reported that its slice step could not find a valid point, with the bracket collapsed to the smallest representable number. The failing point was one of the run’s own live points, stored at a log-likelihood of **-18.480**, above the current threshold of **-19.163**.

Re-evaluated in a fresh process it is **−19.7608540096**, three times identically: the stored value was **1.28 too high**, and the true value is *below* the threshold, so no step from it could ever succeed. The crash was mechanical and deterministic. Re-evaluating **all 800** live points found **four** carrying that same offset — one worker evaluation and its descendants [T2d].

The cause is not new, and that is the point. It had been measured a week earlier in a different configuration: at particular temperatures the oxygen uptake at optimal growth is a **face** of the linear program rather than a vertex, and which point of that face the solver returns is set by its **basis history**. The same signature appears in an earlier crash, in a 0.088-unit variation at identical parameter vectors with *zero* differences in the canonical linear program, and in a 0.028-unit event at a deep-prior draw. **The likelihood evaluated in a persistent worker process is not a deterministic function of its parameters**, and nested sampling — which needs only the *ordering* of likelihood values, but needs that exactly — cannot tolerate it.

Four candidate remedies were then measured on one fixed, hashed set of 22 inputs, 50 evaluations each in one persistent worker, against a fresh-process reference that is bit-identical at all 22. The decision rule was committed before any battery ran and was not revised [T3a]:

scheme	max deviation from the clean value	cost per evaluation	verdict
current	1.815	1.79 s	FAILS
reset the solver before every call	23.277	2.91 s	FAILS
single-solve lexicographic objective	different vertex	1.49 s	FAILS, and disqualified
a fresh model per evaluation	0.000	5.70 s	DETERMINISTIC

Three findings beyond the table, each of which would have been easy to get wrong by intuition:

- **A solver reset per call is worse than doing nothing** — 23.277 against 1.815, at 63 % more cost — and it spreads order-one excursions from three evaluations to 13 of the 19 inputs. It discards the basis but not the *degeneracy*, so each solve re-enters the flat face from a different start. An earlier observation that a reset had restored one value exactly does not generalise, and the matter is now settled.
- **The lexicographic objective is a different model**. It was verified against the current tie-break *before* any determinism verdict, as registered, and it does not return the same vertex at any input: oxygen differs by up to **4.58** and **8.40** mmol gDW^{−1} h^{−1} on the two configurations, growth by up to 2.27 h^{−1} [T3b]. It is faster, and it is not the same target.
- **The corruption attaches to the evaluation, not to the parameter vector**. The point that crashed the run was *clean* over 50 evaluations, while two of its siblings carried the same +1.28 offset. Which input is hit depends on what the worker evaluated before it, so **no parameter vector can be certified clean by a spot check** [T3c]. One configuration is an order of magnitude worse than the other, and it is exactly the one whose oxygen flux sits on a face.

9 A second cause, still open

It would be tidy if this explained everything. It does not, and the test that shows it was registered in advance and reported against interest.

If the two finished runs disagree because their stored likelihoods are corrupted, then their highest-weight samples should carry offsets. **They do not**. The 20 highest-weight posterior samples of each run, re-evaluated in a fresh process, are clean to **2.02 × 10^{−4}** and **5.16 × 10^{−4}**, with nothing above 10^{−4} [T3d]. The defect explains the crash. **It does not explain the disagreement**.

And the disagreement is substantial. Under the revised likelihood the two runs give log evidence **−27.707 ± 0.110** and **−28.047 ± 0.100** — a difference of 0.340 against a combined error of 0.149 — **all 14 physical**

medians differ by more than two Monte-Carlo errors, their leading posterior directions are unrelated, and the appended positive control on a coordinate *proven* to enter nothing is not recovered in either run [T2e]. The honest reading is that the sampler weakness §7 found with no linear program present is still there, and now has to carry this.

The stated limitation of the clean-sample test: those 20 samples cover only **0.3–0.4 %** of the posterior weight. It establishes that the dominant samples are clean, not that the bulk is.

10 What a validated posterior would need, and what it would cost

The protocol was signed before any of these runs, with every threshold naming its calibration source and where that source stops applying (`docs/VALIDATION_PROTOCOL.md @ 9b91420`). Six checks must all pass on five independent runs: an inactive coordinate recovering its exact prior; an appended Beta(3,1) control recovered on its cube with a wrong-prior contrast; consistency of every physical marginal and the leading covariance directions across runs; the prior-rejection fraction with its uncertainty; feasibility-aware predictive checks at every measured temperature including the unresolved-evaluation rate; and evidence reconstruction with independent hash audits. Any single failed check is a failure of the programme.

Three of five reserved seeds are spent. Under the only deterministic scheme, at its measured $3.2\times$ cost, one run of this configuration goes from the **11.2 and 13.4 hours** actually measured to roughly **35–43 hours**, and five sequential runs to **seven to nine days**. The eight remaining configuration fits re-cost from about 100 hours to about 320.

That is the price of the *exact* route. It is not the only route, and §13 treats the alternatives as a menu rather than a plan.

11 What is established

Everything in this section survives the whole of the above. None of it depends on a posterior.

- **The framework and its gates.** One core with seven organism models on it; the Candida gate at **79/79** and the *E. coli* port at **60/60**, run on every change and byte-identical with the options off. The port reproduces the reference implementation’s ten R^2 values to within **0.009** (`reports/P3_gate/gate.md @ fa1fa42`).
- **The mechanisms, implemented and separable.** Measured proteome sectors; overflow metabolism emerging from a total-carbon cap rather than being imposed; the respiratory-chain membrane-area budget; and, from the Candida series, that this budget is expressible in all four Candida models but cannot bind there, because the respiratory chain supplies only **0.02–0.05 %** of the protons ATP synthase consumes (`reports/K4_membrane/`).
- **The audits of the published yeast model.** Our coupling-ion defect class does **not** fire on it; and its temperature-optimum/ceiling asymmetry reproduces independently in their model with their own code, and survives at their **calibrated posterior over 100 models**, where it is stronger and more consistent — with the margin narrowing sharply, so the single-vector figure must not be quoted of their calibrated model [Y2] (`reports/Y1_yeast_audit/`, `reports/Y2_regime_posterior/report.md @ c45fe27`).
- **The predictor validation.** The thermostability predictor resolves differences *between* organisms ($r = +0.76$) and not at all *within* a proteome ($r = -0.05$ on 1,947 measured proteins), which cuts a headline fold-gap from $\sim 79\times$ to $5.4\times$ [**2.5, 10.3**] (`reports/predictor_calibration/`).
- **The likelihood’s machinery**, each piece measured before adoption and each a core option, default off: the parsimonious tie-break, the log-O floor at 1.4216, the `clamp` support form, the removal of the unread coordinate, and the zero-likelihood rule for structural infeasibility.
- **One a-priori prediction, half-successful, correctly labelled.** The *uncalibrated* model predicted an exact-strain growth curve with nothing fitted to growth: it tracked the **shape** and under-predicted the peak **~2.3-fold** (1.04 against 2.40 h^{-1} ; `reports/ecoli_tpc/report.qmd @ c3f66be`). That curve was then consumed by the calibration, so the *calibrated* model has no holdout.

- **That the -4 K stability shift is parameterisation, not biology.** An equally good, equally living fit exists with no shift at all, paid for by the stability scale — but both solutions put the least stable few per cent of enzymes **4–6 °C below** the measured meltome, so the undetermined quantity is the meltome’s low tail, not catalysis (`reports/Y3_tm_shift/report.md @ 1571307`).

12 What is not

- **There is no reproducible posterior for any configuration of any organism in this framework.** Five runs across two versions of the likelihood have met their stopping rules, and no two of them agree. R1 — a posterior two independent runs agree on — is **open**.
- **No interval quoted anywhere in this project is a converged interval.** Every R^2 and every median in the earlier documents carries a not-converged label, and it is real.
- **The predictive check fails on the warm side.** Both finished runs put the measured peak growth (2.076 h^{-1} at 40°C) above their predictive 97.5 % bound (1.62), at four of twelve temperatures [T2e]. The model under-predicts the warm-side peak by about 25 % calibrated — in the same direction as the 2.3-fold uncalibrated gap. Unvalidated runs, so this is a signal to explain, not a result.
- **The mechanism is not discriminated.** The evidence comparison that would distinguish the carbon cap from the membrane-area budget needs one reproducible number per configuration on one medium, and there is not yet one.
- **A second cause of run disagreement is unidentified (§9).**
- **R3 is provisional and R4 is untouched:** the mechanism is not discriminated, and the calibrated model has never been tested out of sample.
- **Two measurements, not models, are missing,** and both are on the critical path: proteome allocation **above 37°C on glucose** — where the thermal ceiling is set and where the measured series stops — and maintenance against temperature from a low-dilution chemostat, the constants for which are currently borrowed.

13 What it would take

A menu, with costs, not a plan. The ordering below is a recommendation and the choice is the PI’s.

1. **Decide the determinism remedy.** Everything else inherits it. The exact route is a fresh model per evaluation at $3.2\times$ cost. Three alternatives have *not* been measured and may be better value: parallelising across runs rather than within them (five single-process runs would fit this machine and cost wall-clock rather than $3.2\times$ the compute); making the tie-break itself state-independent, which would fix the cause rather than the symptom and is the only route that would also help the configurations whose oxygen sits on a face; or accepting the non-reproducibility and abandoning nested sampling for this likelihood.
2. **Then one run under the chosen remedy,** and ask whether the disagreement survives. This is the cheap test of whether the second cause is real, and it is worth one run before committing to five. New seeds must be registered; three of the five reserved ones are spent.
3. **Then the medium comparison** — the same configuration on fully defined M9, changing one thing — with the tie-break instrument run on that medium first, which has never been done.
4. **Then the mechanism comparison,** three configurations on one medium under one likelihood, which is the biological question and is nearly free once one reproducible run exists.
5. **In parallel and independent of all of it:** the two measurements in §12, and the external holdout on evolved lines that would give the calibrated model its first real test.

14 What this document does not license

A validated posterior for one configuration on one medium would license nothing about the others, and there is not one. Nothing here bears on the Candida manuscript’s empirical results, which are not ours to assess. The published-model audits stand on their own evidence and are unaffected by any of the calibration work.

337 And the negative results in §7 and §8 are statements about **this model, this likelihood and this sampler**
338 **configuration** — not about enzyme-constrained models, nested sampling, or calibration in general.